HACKATHON • PROJECT PRESENTATION

EMOTITUNE

Emotion-Based Music Recommendation

Your mood. Your music. Your vibe.

A mood-aware music discovery prototype that lets users choose how they share their mood, personalize song searches, and open their preferred streaming platform.

TEAM INTRODUCTION / 04 MEMBERS



Approach & Problem Statement

A guided workflow reduces the effort of finding music that fits the listener’s current mood.

PROBLEM

Users often spend time searching across platforms for music that fits how they feel.

OUR APPROACH

Offer two mood-input paths: camera-based facial-expression estimation or manual mood selection.

PERSONALIZATION

Combine mood, preferred language, music style and streaming platform into a search query.

USER-CENTRED DESIGN

- Camera use is optional; users can select their mood themselves.
- Users can adjust the suggested mood and choose a preferred genre.
- The result opens a search on YouTube, YouTube Music or Spotify.



Technology Stack

The supplied prototype uses browser-based front-end code and a facial-expression library.

HTML5

Defines the multi-step interface, buttons, mood choices, language selector and result panel.

CSS3

Creates the dark gradient theme, responsive cards, button states, transitions and progress bar.

JAVASCRIPT

Controls navigation, state, input validation, camera flow and search-link generation.

face-api.js

Uses Tiny Face Detector and Face Expression Net to estimate facial expressions in camera mode.

MEDIADEVICES API

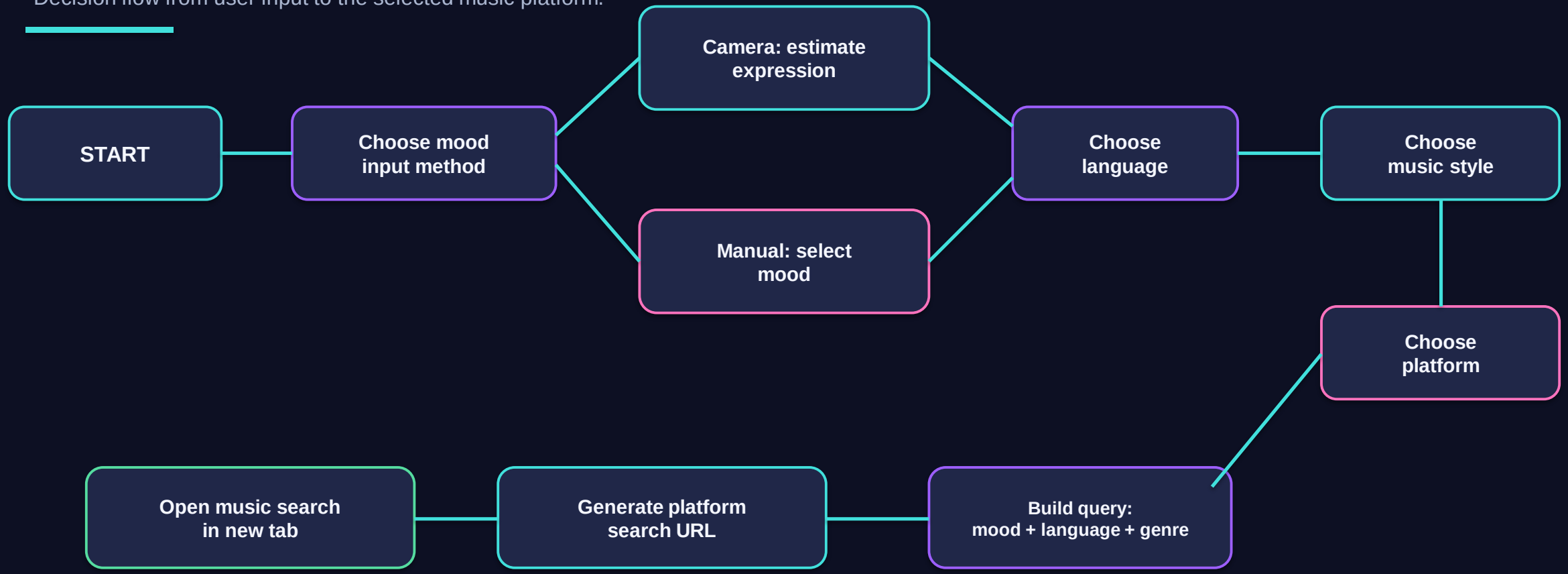
getUserMedia() requests camera access from the browser; the user must grant permission.

MUSIC PLATFORMS

Builds search URLs for YouTube, YouTube Music and Spotify; the selected service opens in a new tab.

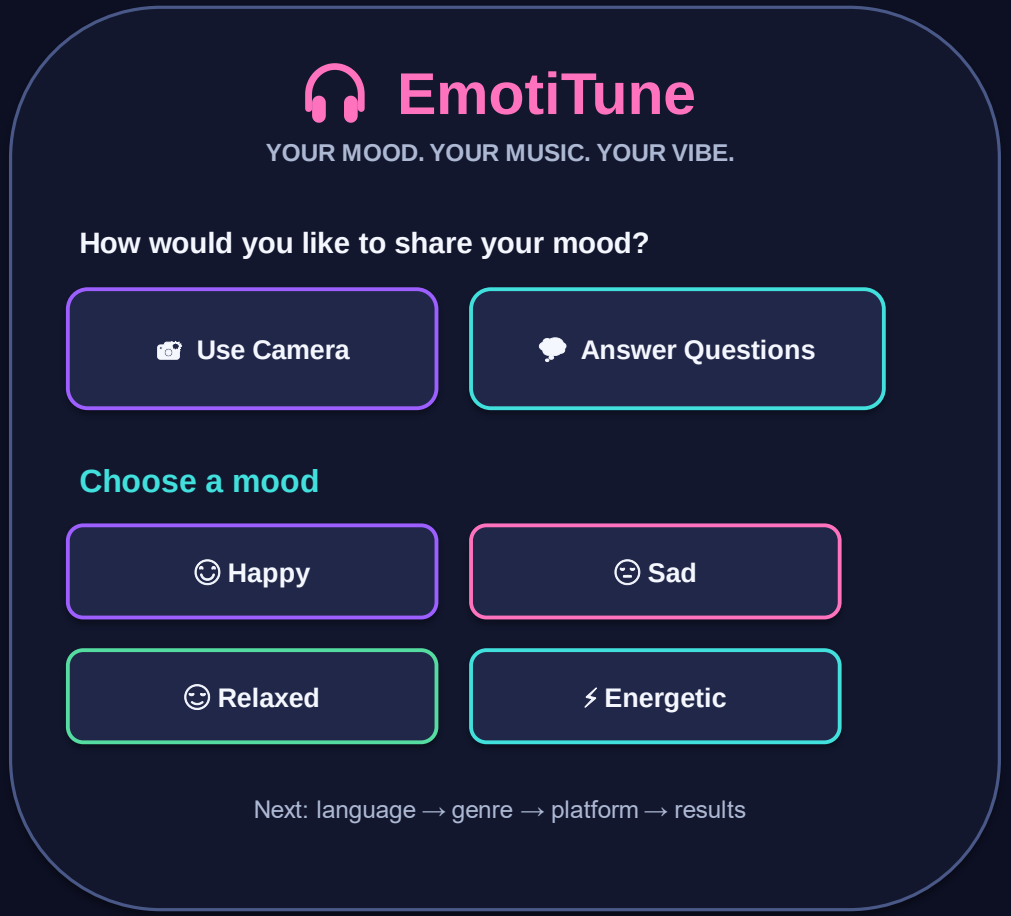
System Flowchart

Decision flow from user input to the selected music platform.



Prototype: What We Built

EmotiTune is a step-by-step website with clear choices and a personalized search result.



PROTOTYPE FEATURES

- 7-step interface with progress indicator.
- Optional camera analysis or manual mood input.
- 8 mood choices and 11 language options.
- Genres include Pop, Lo-fi, Relaxing, Workout and Classical.
- Platform selection: YouTube, YouTube Music or Spotify.
- Summary screen and “Open My Music” action.

DEMO PATH

Start → mood → language → genre → platform → results → open music

References & Future Scope

Technical references for the libraries and browser features used in the prototype.

REFERENCES

OpenAI ChatGPT — AI assistant for code architecture and logic debugging.

Google Gemini — AI research, ideation, and architectural planning.

Google Chrome — Primary browser environment and developer tools for testing and debugging.

face-api.js — Facial expression and face detection library.

Music Streaming Platforms — YouTube , YouTube Music , and Spotify.

FUTURE SCOPE

- Add a Python backend if required for user profiles or recommendation logic.
- Improve recommendations using feedback and listening preferences.
- Integrate official music APIs where permitted by platform terms.
- Test camera permissions, mobile layout and accessibility.
- Make privacy choices clear and let users edit detected mood.

THANK YOU / EMOTITUNE

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